

Lithium-ion Solvation via DFT Cluster Studies

Hasan & Datta (Energy Advances 2025) ¹

Full citation: Rumana Hasan and Dibakar Datta, Computational Study of Li⁺ Solvation Structures in Fluorinated Ether, Non-Fluorinated Ether, and Organic Carbonate-Based Electrolytes at Low and High Salt Concentrations, *Energy Advances* 2025, **4**, 1049–1066 ¹.

Summary: Using MD simulations and DFT calculations, this work explores how Li⁺ is solvated in two weak - solvating ethers (dipropyl ether, DPE, and 2,2-difluoroethyl methyl ether, FEME) vs. a carbonate mixture (ethylene carbonate/diethyl carbonate, EC/DEC). At 1–4 M LiFSI (bis(fluorosulfonyl)imide) in DPE or FEME, large Li⁺-anion aggregates (AGGs) dominate; by contrast, a 1 M LiPF₆ in EC/DEC yields mostly solvent-separated ion pairs (SSIPs). MD shows that FEME (a fluorinated ether) is even weaker at solvating Li⁺ than DPE. The dominant clusters identified by MD were Li⁺(FSI⁻)₃(DPE) and Li⁺(FSI⁻)₃(FEME) (anion-rich), whereas the EC/DEC electrolyte forms Li⁺ with only carbonate solvent coordination (no anion) ². ³ . DFT was then applied to representative clusters (see below) to analyze their electronic structure.

Clusters studied: Representative isolated clusters include Li⁺ with three or four FSI⁻ anions and one ether: e.g. Li⁺(FSI⁻)₃(DPE)₁, Li⁺(FSI⁻)₄(DPE)₁, Li⁺(FSI⁻)₃(FEME)₁, Li⁺(FSI⁻)₄(FEME)₁, as well as anion-free clusters in the carbonate electrolyte (e.g. Li⁺(EC)₁(DEC)₄ and Li⁺(DEC)₅) ⁴. These encompass contact-ion pairs (CIPs, one anion) and aggregates (AGGs, multiple anions) in the primary solvation shell.

DFT analysis: The authors optimized each cluster in vacuum and computed binding/solvation energies and electronic structure. They calculated HOMO/LUMO energies and electrostatic potential (ESP) maps to locate electron density, and performed Bader charge/charge-density-difference analyses to quantify Li→solvent/anions charge transfer ⁵. For example, Table Sx in the SI (not shown) reports Li-solvent binding energies. The DFT results show that Li⁺ is far more positively charged (less bound) in EC/DEC clusters than in ether clusters with anions. In ether clusters Li→FSI charge transfer is significant, consistent with weaker solvent binding.

Key conclusions: The study concludes that in weakly solvating ethers, Li⁺ is sequestered by FSI anions rather than solvent molecules, leading to large Li⁺-anion aggregates. In particular, FEME yields even larger AGGs than DPE; FEME's fluorination makes it an especially weak ligand for Li⁺ ³. The dominant DFT-identified structures are Li⁺(FSI)₃(DPE) and Li⁺(FSI)₃(FEME) ². Overall, the work suggests fluorinated ethers can enforce anion-rich solvation and high-voltage stability, guiding design of “weakly solvating” electrolytes.

Wu et al. (ChemSusChem 2023) ⁶

Full citation: Lan-Qing Wu, Zhe Li, Yong Lu, Jin-Ze Hou, Hao-Qin Han, Qing Zhao, and Jun Chen, Hexacyclic Chelated Lithium Stable Solvates for Highly Reversible Cycling of High-Voltage Lithium Metal Battery, *ChemSusChem* 2023, **16**(18), e202300590 ⁶.

Summary: This study designs new ether solvents that bind Li⁺ in a six-membered chelate ring (hexacyclic structure) instead of the conventional five-membered ring in DME. The authors synthesize 1,3-dimethoxypropane (DMP) and 1,3-diethoxypropane (DEP) as analogues of DME. DFT and spectroscopy confirm that adding a methylene (–CH₂–) unit converts the Li⁺-ether chelate from a five-membered to a six-membered ring ⁷. The six-membered chelates are “weaker” ligands for Li⁺: the Li⁺(DMP)₂ and Li⁺(DEP)₂ clusters are less tightly bound than Li⁺(DME)₂, resulting in improved Li plating/stripping efficiency and stability at high voltage. In practice, a 2.30 M LiFSI/DMP electrolyte enables >90% capacity retention in a Li||NMC811 cell after 184 cycles, under lean electrolyte conditions ⁷.

Clusters studied: The key molecular clusters are Li^+ coordinated by two ether molecules. In particular, $\text{Li}^+(\text{DMP})_2$ and $\text{Li}^+(\text{DEP})_2$ (six-membered chelates) were compared to the conventional $\text{Li}^+(\text{DME})_2$ (five-membered). These clusters model the primary solvation of Li^+ by each solvent. (No anions were included in these DFT clusters – the focus is on Li^+ –solvent complexes.)

DFT analysis: The authors performed DFT (likely B3LYP or similar) geometry optimizations of the $\text{Li}^+(\text{DMP})_2$, $\text{Li}^+(\text{DEP})_2$, and $\text{Li}^+(\text{DME})_2$ clusters. They analyzed the optimized structures to measure ring size and Li–O bond lengths, and used molecular orbital/energy analysis to assess stability. While specifics aren't in the abstract, it notes that “computational” results show the ring transition weakens Li^+ coordination ⁷. Thus, the main DFT outputs are optimized geometries (chelates) and binding energies of Li^+ in each solvent.

Key conclusions: By converting the solvation ring from 5- to 6-membered, DMP/DEP form weaker Li^+ –solvent solvates, which in turn stabilize the interphase and improve cycling. The DFT and experimental data confirm that DMP/DEP make Li^+ “less solvated” (more easily stripped) than DME ⁷. The work highlights that tailoring the Li^+ coordination structure (e.g. via ring size) is a powerful design strategy for high-voltage Li-metal electrolytes.

Choi & Han (Int. J. Mol. Sci. 2024) ⁸ ⁹

Full citation: Jihoon Choi and Young-Kyu Han, High Correlation Between Li^+ Solvation Energy and Li^+ Ionic Conductivity in Lithium Metal Battery Electrolytes, Int. J. Mol. Sci. 2024, **25**(24), 13268 ⁸.

Summary: The authors use first-principles DFT to compute Li^+ solvation energies in model clusters and correlate them with measured ionic conductivity. Previous work often used Li^+ –solvent binding energies in a 1:1 complex, but Choi and Han show that including the counter-anion dramatically improves prediction. They build 1:1:1 clusters (Li^+ + anion + solvent) and calculate the “solvation energy” of Li^+ , defined by the DFT energy of the optimized cluster minus its parts. They find that the calculated Li^+ solvation energy in a $\text{Li}^+:\text{FSI}^-:\text{solvent} = 1:1:1$ cluster correlates extremely well ($R^2 \approx 0.97$) with experimental Li^+ conductivity ⁹. In effect, Li^+ –FSI–solvent complexes (rather than isolated Li–solvent pairs) are needed to capture realistic solvation strength.

Clusters studied: The model clusters are of the form $\text{Li}^+(\text{FSI}^-)(\text{Solv})$, where Solv is a single solvent molecule (various organic carbonates and fluorinated solvents). For example, $\text{Li}^+(\text{FSI}^-)(\text{EC})$, $\text{Li}^+(\text{FSI}^-)(\text{DMC})$, etc. Each cluster contains one Li^+ , one anion (FSI^-) and one solvent molecule ($\text{Li}:\text{P}:\text{F}:\text{S} = 1:1:1$).

DFT analysis: Geometry optimizations and energy calculations were done (B3LYP/6-311G or similar) to obtain cluster energies. The authors computed the Li^+ “solvation energy” (cluster energy minus fragment energies) for each Li^+ –FSI–solvent cluster ⁹. They did not emphasize HOMO/LUMO or charge analysis, but focused on binding/interaction energies. (No mention of Bader or orbital analyses is given; the key data are the relative solvation energies for different solvents.)

Key conclusions: Including the anion in the solvation cluster is crucial: Li^+ solvation energies from the 1:1:1 clusters predict ionic conductivities much better than simple Li^+ –solvent binding energies. The strong correlation ($R^2 \approx 0.97$) shows that Li^+ solvation energy (which includes Li^+ –FSI[−] interaction) is an effective descriptor for solvent solvation ability ⁹. This suggests that future computational screening of Li^+ solvents should model ion pairs (not isolated Li–solvent) to accurately capture transport-relevant interactions.

Sources: Key details above are drawn from the cited papers’ abstracts and sections ¹⁰ ³ ⁴ ⁵ ⁷ ⁹. No additional high-impact cluster-DFT studies beyond these were identified in the specified period. Each reference provides DFT-based insights into Li^+ solvation clusters under the stated conditions.

1 2 3 4 5 10 Computational study of Li⁺ solvation structures in fluorinated ether, non-fluorinated ether, and organic carbonate-based electrolytes at low and high salt concentrations
<https://pubs.rsc.org/en/content/articlepdf/2025/ya/d5ya00154d>

6 7 Hexacyclic Chelated Lithium Stable Solvates for Highly Reversible Cycling of High-Voltage Lithium Metal Battery - PubMed
<https://pubmed.ncbi.nlm.nih.gov/37302979/>

8 9 High Correlation Between Li⁺ Solvation Energy and Li⁺ Ionic Conductivity in Lithium Metal Battery Electrolytes
<https://www.mdpi.com/1422-0067/25/24/13268>